

Building a Globally Distributed Router in Elixir



Orlando
ElixirConf® US

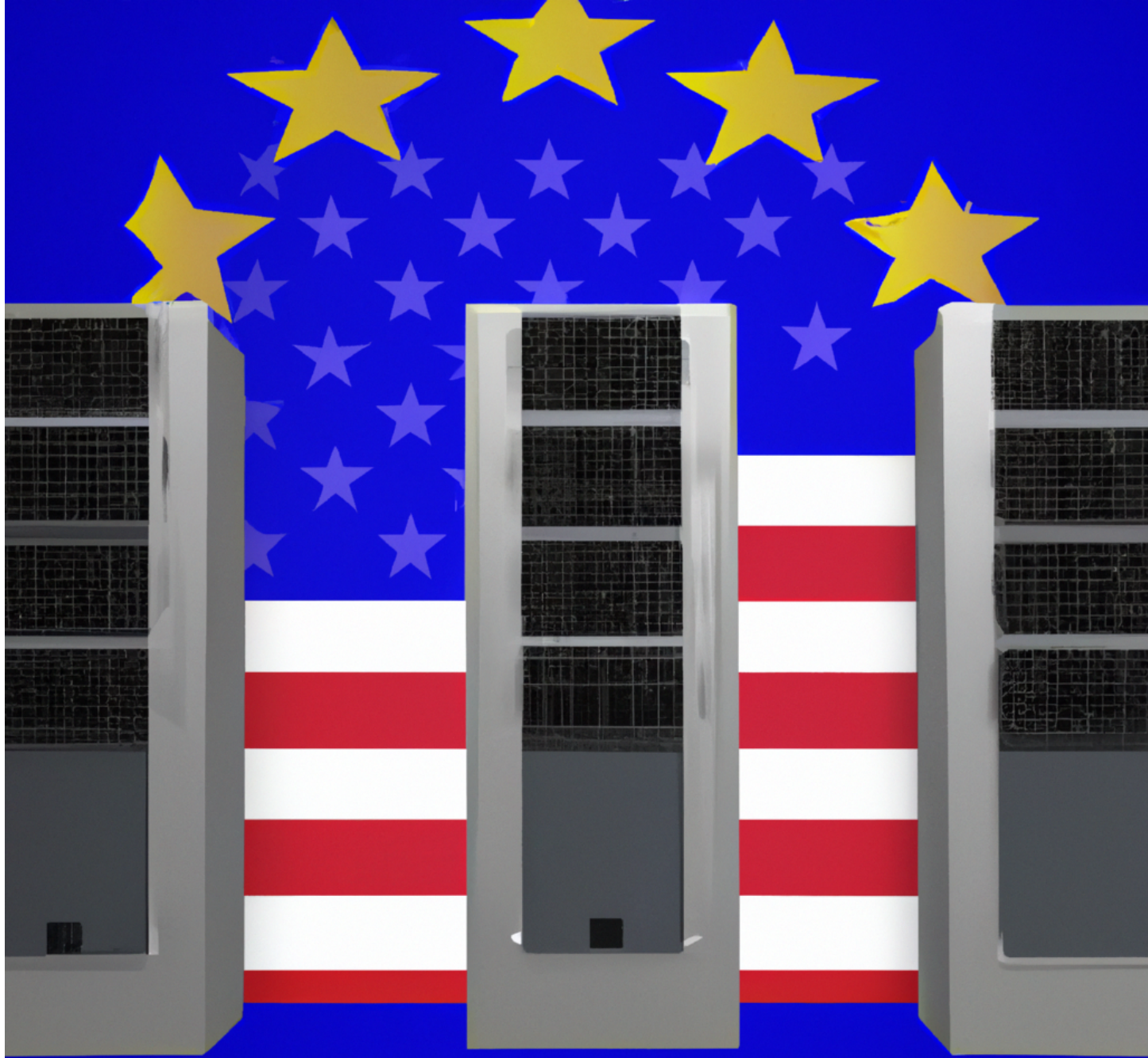
Why did we, and why might you need a router?

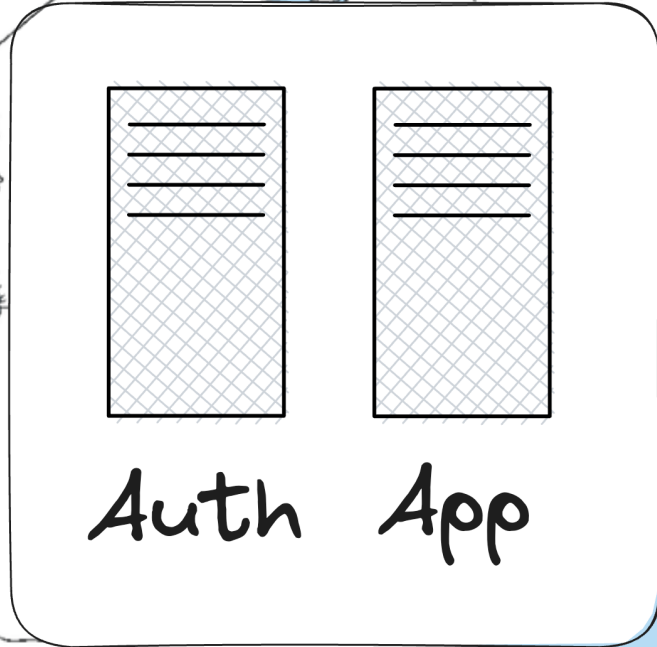


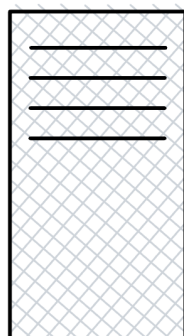
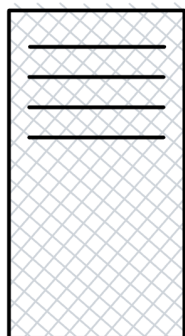
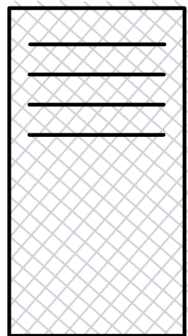
Orlando
ElixirConf US

One server in europe and one server in the USA.

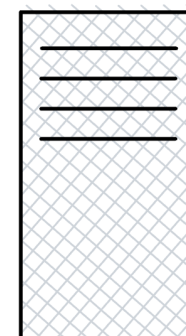
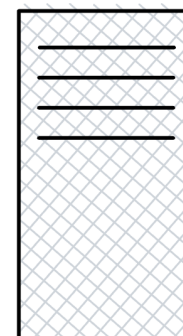
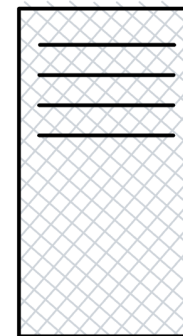








Router Auth App



Router Auth App

Requirements

- Can route a request to the correct region.



Requirements

- Can route a request to the correct region.
- Source of truth for unique email and to which team that email belongs.



Requirements

- Can route a request to the correct region.
- Source of truth for unique email and to which team that email belongs.
- Fast Lookups for routing.

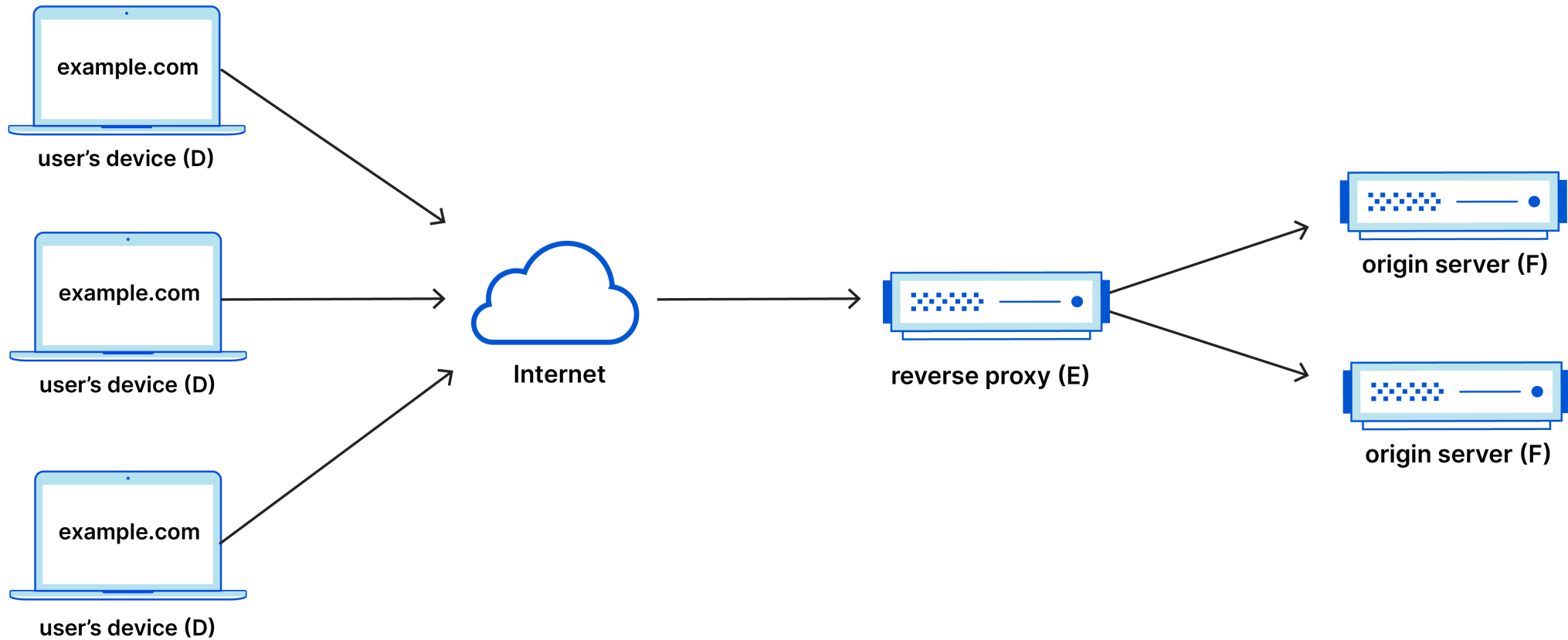


Approach

- Reverse Proxy.



Reverse Proxy Flow



Approach

- Reverse Proxy.
- Single Postgres database in one region, this solves uniqueness.



Approach

- Reverse Proxy.
- Single Postgres database in one region, this solves uniqueness.
- ETS cache in all regions, transported by Kafka, this solves fast lookups.



Elixir Strengths

- Holding on to Connections



Elixir Strengths

- Holding on to Connections
- Plug places the pit of success right in front of you.



Falling in the pit of success, in our `router.ex` file.

```
pipeline :gandalf_proxy do
  plug(RegionDataAdder)
  plug(RegionAdder)
  plug(Parser)
  plug(Proxy)
  plug(SamlNameIdComparer)
  plug(Response)
end
```

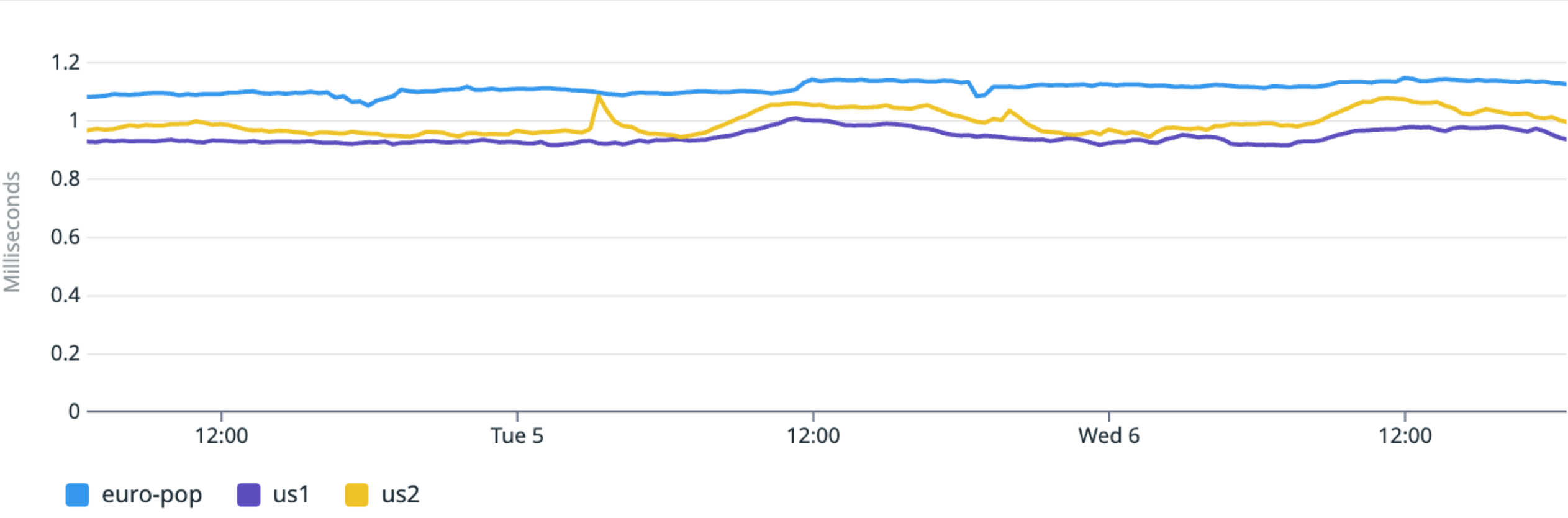


Elixir Strengths

- Holding on to Connections.
- Plug places the pit of success right in front of you.
- Adds minimal latency.



Non-proxy time spent in Randalf



Pitfalls

- Hop-by-hop headers



Pitfalls

- Hop-by-hop headers
- Flexing how much we can do with very little resources



Pitfalls

- Hop-by-hop headers
- Flexing how much we can do with very little resources
- We could not cut any scope, and the project took longer than expected.



Now What?

- Build a proxy, it will be easy.

Now What?

- Build a proxy, it will be easy.
- If you need to build a router, consider how many different types of requests you need to route.